

D. Self-Study

Sr. No.	Topic	Unit
1	Encourage independent and lifelong learning	Unit III
2	Develop research and analytical capabilities	Unit V
3	Explore advanced and emerging security paradigms	Unit IV & V
4	Bridge theory with real-world system and network security practices	All Units

Information Security**A. OVERVIEW OF THE COURSE**

Name of Course	Information Security
Offering Department	Information Technology

Overview:

This course provides a comprehensive understanding of mathematical foundations, cryptographic techniques, and system-level security mechanisms required to protect information assets. It covers number theory, symmetric and public-key cryptography, cryptographic hash functions, digital signatures, and network and system security mechanisms. Aligned with NEP-2020, the course emphasizes analytical reasoning, secure system design, hands-on experimentation, and real-world security practices relevant to modern digital infrastructures.

B. Syllabus

Course No	Title of Course	Credits	L-T-P	Pre- Requisites
CYITC102	Information Security	4	3L-0T-2P	None
COURSE OUTCOMES: CO1 – Apply number theory concepts to understand cryptographic primitives CO2 – Design and analyse traditional and modern symmetric cryptographic algorithms CO3 – Implement and evaluate public-key cryptographic algorithms and their security CO4 – Understand integrity, authentication, and digital signature mechanisms CO5 – Analyse network and system security threats and deploy suitable protection mechanisms				
Unit No	Topics			
Unit 1	Number Theory Groups, Rings, Finite Fields, Modular Arithmetic, Matrices, Linear Congruence, Efficient Algorithms for Modular Arithmetic, Euler's Totient Function, Euler's Theorem, Fermat's Little Theorem, Primality Testing, Prime Factorization, Chinese Remainder theorem, Quadratic Congruence, Euler's Criteria			
Unit 2	Traditional and Modern Symmetric Ciphers Traditional Ciphers: Affine Cipher, Autokey Cipher, Playfair Cipher, Vigenere Cipher, Hill Cipher, Transposition Ciphers, Stream and Block Ciphers, Modern Ciphers: Components of Modern Block Ciphers, Product ciphers, Diffusion, Confusion, Fiestel and Non-Fiestel Ciphers, Linear and Differential Cryptanalysis, Modern Stream Cipher, Blowfish, DES, AES			
Unit 3	Public Key Cryptography RSA, Knapsack, Rabin, ElGamal, Elliptic Curve Cryptography, Diffie Hellman Key Exchange algorithm, Lattice based Cryptography			

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Annexure-II

Unit 4	Cryptographic Hash Function and Digital Signatures: Hash functions and its properties, SHA-512, Digital Signature, Attacks on Digital Signatures, Digital Signatures Schemes like RSA, ElGamal, Schnorr, DSS, ECC, Variations and Applications.
Unit 5	Network and System Security Network Security: Two Modes of IPSec, Two Security Protocols AH and ESP, Security Association, Security Policy, Internet Key Exchange System Security: Buffer Overflow and Malicious Software, Malicious Programs, Worms, Viruses, Intrusion Detection System, Firewalls and its Types, Virtual Private Networks
SUGGESTED READINGS: 1. B. A. Forouzan, "Cryptography & Network Security", Tata Mc Graw Hill. 2. J.Katz, Y.Lindell, Introduction to Modern Cryptography, Chapman Hall. 3. Wen Bo Mao, Modern cryptography - Theory and practice, Prentice Hall. 4. Douglas Stinson, "Cryptography Theory and Practice", Second Edition, Chapman & Hall/CRC. 5. W. Stallings, "Cryptography and Network Security", Pearson Education	

C. Theory Lecture Plan

C. Theory Lecture Plan			
Sr. No.	CONTENT	NUMBER OF LECTURES	Unit
1	Groups, Rings	2	Unit I (10)
2	Finite Fields, Modular Arithmetic	1	
3	Matrices, Linear Congruence	1	
4	Efficient Algorithms for Modular Arithmetic	1	
5	Euler's Totient Function, Euler's Theorem, Fermat's Little Theorem	1	
6	Advanced Number Theory: Primality Testing, Prime Factorization	2	
7	Chinese Remainder theorem	1	
8	Quadratic Congruence, Euler's Criteria	1	
9	Affine Cipher, Autokey Cipher,	1	Unit II (10)
10	Playfair Cipher, Vigenere Cipher	1	
11	Hill Cipher	1	
12	Transposition Ciphers, Stream and Block Ciphers	1	
13	Modern Block ciphers, Components of a block cipher	1	
14	Product ciphers, Diffusion, Confusion, Fiestel and Non-Fiestel Ciphers	1	
15	Linear and Differential Cryptanalysis, Modern Stream Cipher,	1	
16	Blowfish	1	
17	DES	1	
18	AES	1	
MID SEMESTER EXAMS			
19	RSA	1	Unit III (7)
20	Knapsack	1	
21	Rabin	1	
22	ElGamal	1	
23	Elliptic Curve	1	
24	Diffie Hellman Key Exchange algorithm,	1	
25	Lattice based Cryptography	1	

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Annexure-II

26	Hash functions and its properties	1	Unit IV (6)
27	SHA-512	1	
28	Digital Signature, Attacks on Digital Signatures, RSA	1	
29	ElGamal,	1	
30	Schnorr, DSS	1	
31	ECC, Variations and Applications	1	
32	Network Security: Two Modes of IPSec, Two Security Protocols AH and ESP	1	Unit V (7)
33	Security Association, Security Policy	1	
34	Internet Key Exchange	1	
35	Buffer Overflow and Malicious Software, Malicious Programs, Worms, Viruses	2	
36	Intrusion Detection System, Firewalls and its Types, Virtual Private Networks	2	

D. Lab Experiments

No.	Name of Experiment
1	Implementation of modular arithmetic and primality testing
2	Implementation of classical substitution and transposition ciphers
3	Implementation and analysis of AES encryption and decryption
4	Implementation of RSA public-key cryptosystem
5	Implementation of Diffie–Hellman key exchange
6	Implementation of cryptographic hash functions (SHA-512)
7	Digital signature generation and verification
8	Simulation of network security using IPSec
9	Firewall and intrusion detection system configuration

E. Self-Study

Sr. No.	Topic	
1.	Cryptanalysis of classical and modern ciphers	Unit I
2.	Case study on real-world security breaches	Unit II
3.	Post-quantum cryptography overview	Unit V
4.	Any topic suggested by the course instructor	Unit V

Cyber Laws**A. Overview**

Name of Course	Cyber Laws
Offering Department	Information Technology

The Cyber Laws syllabus provides an understanding of the legal and regulatory framework governing cyberspace, focusing on the prevention, detection, and prosecution of cybercrimes. It covers key legislations such as the Information Technology Act, cyber offences and penalties, digital evidence, data protection and privacy laws, intellectual property issues in the digital domain, and emerging challenges like cyber terrorism and social media misuse. The syllabus equips learners with legal awareness to ensure compliance, promote cyber security, and address legal issues arising from the use of computers, networks, and the internet

B. Syllabus

Course No.	Title of the Course	Credits	Course Structure	Pre-Requisite
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